Adaptation-Aware Medical Core Pruning for MedSAM

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Abstract

We present a method section draft for structured pruning of MedSAM. The main idea is simple. MedSAM is initialized from SAM and then adapted to medical images. We use this adaptation path to identify the parts of MedSAM that truly form a *medical core*. Our formulation combines a standard pruning view, namely budget-constrained medical risk minimization, with a new reset-based score that measures whether a structure is still needed after medical adaptation. We keep the notation explicit and the writing simple so that this version can be used for teaching and internal discussion.

1 Method

Our goal is to compress MedSAM while preserving segmentation quality on medical images. We start from two facts. First, MedSAM keeps the main architecture of SAM, namely an image encoder, a prompt encoder, and a mask decoder [Kirillov et al., 2023, Ma et al., 2024]. Second, MedSAM is obtained by adapting SAM to medical images, rather than by training a new model from scratch [Ma et al., 2024]. Therefore, the compressed model should not only keep structures that are useful on medical data, but also keep structures that became specialized during medical adaptation. We call the remaining structure the *medical core*.

The method section is organized as follows. We first define the model, the data, and the pruning objective. We then introduce a dual-intervention score for structured pruning. After that, we derive a practical second-order approximation and describe the optimization procedure. We end with a short discussion of which parts are theory-driven and which parts are still practical approximations.

1.1 MedSAM setup and notation

Each training or calibration sample has three parts, namely the image, the box prompt, and the target mask. We write one sample as

$$\mathbf{z} = (\mathbf{I}, \mathbf{b}, \mathbf{M}), \tag{1}$$

where $\mathbf{I} \in \mathbb{R}^{3\times H\times W}$ is the preprocessed medical image, $\mathbf{b} \in \mathbb{R}^4$ is a bounding box prompt, and $\mathbf{M} \in \{0,1\}^{H\times W}$ is the binary target mask.

MedSAM follows the SAM design. Let $E_{\boldsymbol{\theta}_E}$ be the image encoder, let $P_{\boldsymbol{\phi}}$ be the prompt encoder, and let $D_{\boldsymbol{\theta}_D}$ be the mask decoder. In the original MedSAM training recipe, the prompt encoder is frozen and the model mainly uses box prompts [Ma et al., 2024]. Therefore, the learnable parameters that matter for pruning are

$$\boldsymbol{\theta} = (\boldsymbol{\theta}_E, \boldsymbol{\theta}_D), \tag{2}$$

while $\boldsymbol{\phi}$ is fixed.

The model output is written as

$$f_{\boldsymbol{\theta},\boldsymbol{\phi}}(\mathbf{I}, \mathbf{b}) = D_{\boldsymbol{\theta}_D}\big(E_{\boldsymbol{\theta}_E}(\mathbf{I}), P_{\boldsymbol{\phi}}(\mathbf{b})\big), \tag{3}$$

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where $f_{\boldsymbol{\theta},\boldsymbol{\phi}}(\mathbf{I},\mathbf{b}) \in \mathbb{R}^{H\times W}$ denotes the output mask logits. We denote the predicted mask probability by

$$\hat{\mathbf{M}} = \sigma\big(f_{\boldsymbol{\theta},\boldsymbol{\phi}}(\mathbf{I},\mathbf{b})\big), \tag{4}$$

where $\sigma(\cdot)$ is the element-wise sigmoid.

Following MedSAM, we use the sum of a Dice loss and a binary cross-entropy loss [Ma et al., 2024]. For one sample $\mathbf{z}$, the loss is

$$\ell(\mathbf{z};\boldsymbol{\theta}) = \ell_{\mathrm{Dice}}(\hat{\mathbf{M}},\mathbf{M}) + \ell_{\mathrm{BCE}}\big(f_{\boldsymbol{\theta},\boldsymbol{\phi}}(\mathbf{I},\mathbf{b}),\mathbf{M}\big). \tag{5}$$

Let $\mathcal{D}_{\mathrm{cal}} = \{\mathbf{z}_n\}_{n=1}^{N}$ be a small medical calibration set. We define the empirical medical risk as

$$\widehat{R}_{\mathrm{med}}(\boldsymbol{\theta}) = \frac{1}{N}\sum_{n=1}^{N} \ell(\mathbf{z}_n;\boldsymbol{\theta}). \tag{6}$$

This calibration set does not need to be large. Its role is to estimate which structures are important on medical data.

Our method uses two checkpoints. We denote by $\boldsymbol{\theta}^S$ the SAM checkpoint before medical adaptation, and by $\boldsymbol{\theta}^M$ the MedSAM checkpoint after medical adaptation. The starting point of pruning is

$$\boldsymbol{\theta}^{(0)} = \boldsymbol{\theta}^M. \tag{7}$$

The reference checkpoint $\boldsymbol{\theta}^S$ is only used to measure how much each structure changed from SAM to MedSAM.

1.2 Structured pruning objective

We do not prune individual weights. Instead, we prune structured groups that can be removed from the network in practice. Let

$$\mathcal{G} = \{g_1, g_2, \ldots, g_K\} \tag{8}$$

be the set of all pruning groups. A group can be one attention head, one group of MLP hidden channels, or one group of decoder channels. Concrete examples are given in Appendix B.

We assign one binary variable to each group and collect them in a vector

$$\mathbf{s} = (s_{g_1}, s_{g_2}, \ldots, s_{g_K}) \in \{0,1\}^K. \tag{9}$$

If $s_g = 1$, group g is active. If $s_g = 0$, group g is pruned. For any parameter index i, let $g(i)$ denote the group that contains parameter θ_i. We define group-wise masking by

$$(\mathbf{s} \odot_G \boldsymbol{\theta})_i = s_{g(i)}\theta_i. \tag{10}$$

Each group has a nonnegative cost c_g, which can represent parameter count, FLOPs, or measured latency gain. The total cost of the active model is

$$C(\mathbf{s}) = \sum_{g\in\mathcal{G}} s_g c_g. \tag{11}$$

Given a target budget B, the ideal pruning problem is

$$\min_{\boldsymbol{\theta},\,\mathbf{s}} \quad \widehat{R}_{\mathrm{med}}(\mathbf{s} \odot_G \boldsymbol{\theta}) \tag{12}$$

$$\text{s.t.} \quad C(\mathbf{s}) \le B, \qquad \mathbf{s} \in \{0,1\}^K. \tag{13}$$

This formulation is simple and standard. The challenge is that it is a combinatorial optimization problem. We therefore need a group score that tells us which structures are the safest to remove.

1.3 Dual-intervention medical-core score

The key question is not only whether a group is useful now, but also whether it became specialized during medical adaptation. To answer both questions, we define two interventions for each group g.

The first intervention is a zero intervention. It removes the current contribution of group g by setting its parameters to zero. Formally, for a parameter vector $\boldsymbol{\theta}$, we define

$$\left[\mathcal{T}_g^0(\boldsymbol{\theta})\right]_i = \begin{cases} 0, & i \in g, \\ \theta_i, & i \notin g. \end{cases} \tag{14}$$

Its exact medical risk increase is

$$\Delta_g^{\text{zero}}(\boldsymbol{\theta}, \mathbf{s}) = \widehat{R}_{\text{med}}\big(\mathbf{s} \odot_G \mathcal{T}_g^0(\boldsymbol{\theta})\big) - \widehat{R}_{\text{med}}\big(\mathbf{s} \odot_G \boldsymbol{\theta}\big). \tag{15}$$

If this value is small, group g is not very important for the current MedSAM model.

The second intervention is a reset intervention. It keeps the group active, but resets its parameters from the current MedSAM values back to the original SAM values. We define

$$\left[\mathcal{T}_g^S(\boldsymbol{\theta})\right]_i = \begin{cases} \theta_i^S, & i \in g, \\ \theta_i, & i \notin g. \end{cases} \tag{16}$$

Its exact medical risk increase is

$$\Delta_g^{\text{reset}}(\boldsymbol{\theta}, \mathbf{s}) = \widehat{R}_{\text{med}}\big(\mathbf{s} \odot_G \mathcal{T}_g^S(\boldsymbol{\theta})\big) - \widehat{R}_{\text{med}}\big(\mathbf{s} \odot_G \boldsymbol{\theta}\big). \tag{17}$$

If this value is small, the medical loss barely changes even after we restore this group to the SAM state. In that case, this group did not become strongly specialized during medical adaptation.

We combine the two effects into one group score:

$$Q_g(\boldsymbol{\theta}, \mathbf{s}) = \alpha\, \Delta_g^{\text{zero}}(\boldsymbol{\theta}, \mathbf{s}) + (1 - \alpha)\, \Delta_g^{\text{reset}}(\boldsymbol{\theta}, \mathbf{s}), \tag{18}$$

where $\alpha \in [0, 1]$ controls the balance between current usefulness and adaptation-specific usefulness.

This definition is the core of the method. A group receives a small score when two conditions hold at the same time. First, removing it does not increase the medical risk much. Second, restoring it to the SAM state also does not increase the medical risk much. Such a group is a natural pruning candidate because it is weak both as a current medical structure and as an adaptation-specific medical structure.

In practice, we also normalize the score by the pruning benefit of each group. Let c_g be the cost defined in Eq. (11). We use the cost-aware priority

$$P_g(\boldsymbol{\theta}, \mathbf{s}) = \frac{Q_g(\boldsymbol{\theta}, \mathbf{s})}{(c_g + \varepsilon)^\tau}, \tag{19}$$

where $\varepsilon > 0$ avoids numerical issues and $\tau \geq 0$ controls how strongly the score favors groups with larger pruning benefit. We prune groups with the smallest priorities first.

1.4 Second-order approximation

The exact scores in Eqs. (15) and (17) are clean, but they are too expensive to evaluate for every group. We therefore use a standard local approximation from second-order pruning. The idea goes back to Optimal Brain Damage and later Fisher-based structured pruning [LeCun et al., 1990, Liu et al., 2021].

We expand the empirical medical risk around the current parameter vector $\boldsymbol{\theta}$:

$$\widehat{R}_{\text{med}}(\boldsymbol{\theta} + \boldsymbol{\delta}) \approx \widehat{R}_{\text{med}}(\boldsymbol{\theta}) + \nabla \widehat{R}_{\text{med}}(\boldsymbol{\theta})^\top \boldsymbol{\delta} + \frac{1}{2} \boldsymbol{\delta}^\top \mathbf{H}(\boldsymbol{\theta}) \boldsymbol{\delta}, \tag{20}$$

where $\mathbf{H}(\boldsymbol{\theta}) = \nabla^2 \widehat{R}_{\text{med}}(\boldsymbol{\theta})$ is the Hessian. After medical fine-tuning, the current parameters are usually close to a local stationary point, so the first-order term is small. Under this approximation, the two interventions have simple forms.

For the zero intervention, the local parameter change on group g is $\boldsymbol{\delta}_g^{\text{zero}} = -\boldsymbol{\theta}_g$. Therefore,

$$\Delta_g^{\text{zero}}(\boldsymbol{\theta}, \mathbf{s}) \approx \frac{1}{2} \boldsymbol{\theta}_g^\top \mathbf{H}_g(\boldsymbol{\theta})\, \boldsymbol{\theta}_g, \tag{21}$$

where $\mathbf{H}_g(\boldsymbol{\theta})$ is the local Hessian block for group g.

For the reset intervention, the local parameter change is

$$\boldsymbol{\delta}_g^{\text{reset}} = \boldsymbol{\theta}_g^S - \boldsymbol{\theta}_g. \tag{22}$$

Thus,

$$\Delta_g^{\text{reset}}(\boldsymbol{\theta}, \mathbf{s}) \approx \frac{1}{2}(\boldsymbol{\theta}_g - \boldsymbol{\theta}_g^S)^\top \mathbf{H}_g(\boldsymbol{\theta})(\boldsymbol{\theta}_g - \boldsymbol{\theta}_g^S). \tag{23}$$

To make the method practical, we replace the Hessian by a diagonal Fisher approximation. For each parameter θ_i, we define

$$F_i(\boldsymbol{\theta}, \mathbf{s}) = \frac{1}{N} \sum_{n=1}^{N} \left(\frac{\partial \ell(\mathbf{z}_n; \mathbf{s} \odot_G \boldsymbol{\theta})}{\partial \theta_i} \right)^2. \tag{24}$$

Then the two approximate group costs become

$$\widehat{\Delta}_g^{\text{zero}}(\boldsymbol{\theta}, \mathbf{s}) = \frac{1}{2} \sum_{i \in g} F_i(\boldsymbol{\theta}, \mathbf{s})\, \theta_i^2, \tag{25}$$

$$\widehat{\Delta}_g^{\text{reset}}(\boldsymbol{\theta}, \mathbf{s}) = \frac{1}{2} \sum_{i \in g} F_i(\boldsymbol{\theta}, \mathbf{s})\, (\theta_i - \theta_i^S)^2. \tag{26}$$

The practical score is then

$$\widehat{Q}_g(\boldsymbol{\theta}, \mathbf{s}) = \alpha\, \widehat{\Delta}_g^{\text{zero}}(\boldsymbol{\theta}, \mathbf{s}) + (1 - \alpha)\, \widehat{\Delta}_g^{\text{reset}}(\boldsymbol{\theta}, \mathbf{s}), \tag{27}$$

$$\widehat{P}_g(\boldsymbol{\theta}, \mathbf{s}) = \frac{\widehat{Q}_g(\boldsymbol{\theta}, \mathbf{s})}{(c_g + \varepsilon)^\tau}. \tag{28}$$

We use $\widehat{P}_g$ to rank all active groups.

The two terms in Eq. (27) play different roles. The first term favors groups that are unimportant for the current MedSAM model. The second term favors groups that changed little, or changed in directions that are unimportant on medical loss. When both terms are small, the group is unlikely to belong to the medical core.

1.5 Distribution-aware extension

The formulation above already uses medical data through the calibration risk in Eq. (6). However, medical data often come from several sub-distributions, such as different modalities, hospitals, scanners, or box-prompt regimes. A single average score may hide structures that are critical for one subset but not for others.

To address this issue, we split the calibration set into R subsets,

$$\mathcal{D}_{\text{cal}} = \bigcup_{r=1}^{R} \mathcal{D}_{\text{cal}}^{(r)}. \tag{29}$$

For each subset r, we compute a subset-specific score $\widehat{Q}_g^{(r)}$ by using the same formulas as in Eqs. (25) to (27), but only on $\mathcal{D}_{\text{cal}}^{(r)}$.

We then define a distribution-aware score as

$$\widehat{Q}_g^{\text{dist}} = \sum_{r=1}^{R} \pi_r\, \widehat{Q}_g^{(r)} + \beta\, \mathrm{Var}_{r=1,\ldots,R}\big(\widehat{Q}_g^{(r)}\big), \tag{30}$$

where $\pi_r \geq 0$ and $\sum_{r=1}^{R} \pi_r = 1$ are user-defined subset weights, and $\beta \geq 0$ controls the penalty on cross-subset instability.

This extension is easy to interpret. The weighted average term keeps structures that are important on the expected medical mixture. The variance term keeps structures whose importance is highly uneven across subsets. The final priority score becomes

$$\widehat{P}_g^{\text{dist}} = \frac{\widehat{Q}_g^{\text{dist}}}{(c_g + \varepsilon)^\tau}. \tag{31}$$

When $\beta = 0$, the method reduces to the average-case version.

1.6 Optimization

We solve the pruning problem in an iterative prune-and-recover manner. We begin from the full MedSAM checkpoint $\boldsymbol{\theta}^{(0)} = \boldsymbol{\theta}^{M}$ and the all-one group mask $\mathbf{s}^{(0)}$. At each stage t, we estimate the diagonal Fisher scores on the current masked model $(\mathbf{s}^{(t)} \odot_G \boldsymbol{\theta}^{(t)})$ by a forward and backward pass over the medical calibration set.

After that, we compute the group priorities in Eq. (28), or Eq. (31) for the distribution-aware variant. We then prune a small fraction of active groups with the lowest priorities. The group mask is updated, and a short recovery fine-tuning stage follows on medical training data. During recovery, the pruned groups stay inactive and only the remaining parameters are updated. This process repeats until the target budget B is reached. Detailed pseudocode is given in Appendix A.

This optimization is simple and practical. The calibration pass gives a local estimate of medical importance. The recovery step corrects the approximation error from one pruning stage before the next stage begins. In practice, the prune ratio per stage should be small enough to avoid large jumps in the loss surface.

1.7 Discussion of theory and heuristics

It is useful to state clearly what is theory-driven and what is heuristic. The overall problem in Eq. (12) is a standard budget-constrained empirical risk minimization problem. The local approximation in Eqs. (21) and (23) follows the same second-order logic used in classical pruning work and in later Fisher-based structured pruning [LeCun et al., 1990, Liu et al., 2021]. From this view, our method is not a purely heuristic ranking rule.

The new part is the reset intervention. This term is a modeling choice that uses the adaptation path from SAM to MedSAM. It asks whether a group has become specifically useful for medical segmentation, instead of only asking whether the group is useful at the current parameters. This idea is related in spirit to the observation that pretrained models should be pruned with adaptation in mind, rather than only with weight magnitude in mind [Sanh et al., 2020]. However, the exact form of our reset score is still task-driven. It does not come from a closed-form optimality theorem.

Therefore, the best description is the following. The method has a clear theoretical core, namely constrained medical risk minimization with a local second-order approximation. At the same time, the practical algorithm still contains approximations and design choices. These include diagonal Fisher instead of the full Hessian, the definition of structured groups, the hyperparameters α, β, and τ, and the greedy prune-and-recover schedule. In this sense, the method is theory-guided, but not theory-complete.

A Detailed pseudocode

Algorithm 1 gives the detailed procedure. The same algorithm can be used with the distribution-aware score by replacing $\widehat{Q}_g$ with $\widehat{Q}_g^{\mathrm{dist}}$.

B Examples of structured groups in MedSAM

This appendix gives two common examples of group definitions in a ViT-style MedSAM encoder.

For an attention head, let $\mathcal{I}_{\ell,h}$ be the index set of head h in transformer block ℓ. A coupled attention-head group can be written as

$$g_{\ell,h}^{\mathrm{att}} = \Big\{\mathbf{W}_Q^{(\ell)}[:, \mathcal{I}_{\ell,h}], \mathbf{W}_K^{(\ell)}[:, \mathcal{I}_{\ell,h}], \mathbf{W}_V^{(\ell)}[:, \mathcal{I}_{\ell,h}], \mathbf{W}_O^{(\ell)}[\mathcal{I}_{\ell,h}, :]\Big\}. \tag{32}$$

All four tensors must be pruned together. Otherwise, the head dimensions will no longer match.

For an MLP group, let $\mathcal{J}_{\ell,r}$ be a subset of hidden-channel indices in the MLP of block ℓ. A coupled MLP group can be written as

$$g_{\ell,r}^{\mathrm{mlp}} = \Big\{\mathbf{W}_1^{(\ell)}[:, \mathcal{J}_{\ell,r}], \mathbf{W}_2^{(\ell)}[\mathcal{J}_{\ell,r}, :]\Big\}. \tag{33}$$

Again, both tensors must be pruned together. This coupling is important for real structural compression, not only for sparse masking.

Algorithm 1 Adaptation-aware medical core pruning for MedSAM

Require: SAM checkpoint $\boldsymbol{\theta}^S$; MedSAM checkpoint $\boldsymbol{\theta}^M$; fixed prompt encoder $\boldsymbol{\phi}$; medical calibration set $\mathcal{D}_{\mathrm{cal}}$; medical recovery set $\mathcal{D}_{\mathrm{rec}}$; group set $\mathcal{G}$; group costs $\{c_g\}_{g\in\mathcal{G}}$; target budget B; trade-off parameters α, τ; stability constant ε; pruning ratio per stage ρ; recovery steps T_{rec}.

Ensure: Compact pruned model $\boldsymbol{\theta}^\star$.

Initialize $\boldsymbol{\theta}^{(0)} \leftarrow \boldsymbol{\theta}^M$.
Initialize $\mathbf{s}^{(0)}$ as the all-one mask over $\mathcal{G}$.
Set stage index $t \leftarrow 0$.
while $C(\mathbf{s}^{(t)}) > B$ **do**
  Set all Fisher statistics $F_i \leftarrow 0$ for active parameters.
  for all mini-batches $\mathcal{B} \subset \mathcal{D}_{\mathrm{cal}}$ **do**
    Run a forward pass of the current masked model $\hat{\mathbf{Y}} \leftarrow f_{\mathbf{s}^{(t)}\odot_G\boldsymbol{\theta}^{(t)},\boldsymbol{\phi}}(\mathbf{I},\mathbf{b})$ for all $(\mathbf{I},\mathbf{b},\mathbf{M}) \in \mathcal{B}$.
    Compute the MedSAM loss on the batch: $L \leftarrow \sum_{(\mathbf{I},\mathbf{b},\mathbf{M})\in\mathcal{B}} \ell\big((\mathbf{I},\mathbf{b},\mathbf{M}); \mathbf{s}^{(t)} \odot_G \boldsymbol{\theta}^{(t)}\big)$.
    Backpropagate L and accumulate squared gradients: $F_i \leftarrow F_i + (\partial L/\partial\theta_i)^2/N$ for all active i.
  end for
  for all active groups $g \in \mathcal{G}$ with $s_g^{(t)} = 1$ **do**
    Compute the zero score: $\widehat{\Delta}_g^{\mathrm{zero}} \leftarrow \frac{1}{2}\sum_{i\in g} F_i\,(\theta_i^{(t)})^2$.
    Compute the reset score: $\widehat{\Delta}_g^{\mathrm{reset}} \leftarrow \frac{1}{2}\sum_{i\in g} F_i\,(\theta_i^{(t)} - \theta_i^S)^2$.
    Combine the two terms: $\widehat{Q}_g \leftarrow \alpha\,\widehat{\Delta}_g^{\mathrm{zero}} + (1-\alpha)\,\widehat{\Delta}_g^{\mathrm{reset}}$.
    Compute the cost-aware priority: $\widehat{P}_g \leftarrow \widehat{Q}_g/(c_g+\varepsilon)^\tau$.
  end for
  Rank all active groups by $\widehat{P}_g$ in ascending order.
  Select the lowest-score groups that correspond to a fraction ρ of the currently active groups.
  Update the group mask by setting the selected group indicators to zero.
  Recover the remaining model by fine-tuning on $\mathcal{D}_{\mathrm{rec}}$ for T_{rec} steps while keeping the pruned groups inactive.
  Optionally compact the current network by physically removing the pruned heads and channels.
  Set $t \leftarrow t+1$.
end while
Return the compact masked model as $\boldsymbol{\theta}^\star$.

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